Digital Logic System

Assignment 2

1. What is the difference between Maxterm and Minterm? Show that the dual of the exclusive OR is equal to its compliment.
2. What are logic gates? Name the three elementary or basic gates and explain about them.
3. What are postulates of Boolean algebra?
4. Show NAND and NOR gates are universal gates.
5. What is Boolean algebra? State and prove the Demorgan's therorems with three variables.
6. State the distributive law of Boolean algebra.
7. What do you mean by universal gate? Show that NAND and NOR gates are universal gates.
8. Define canonical form. Convert the following given Boolean functions in to standard minterm and maxterm.
   i. $F = A + BC$
9. Simplify the following Boolean functions in SOP and POS form.
   i. $F(A, B, C, D) = \Sigma(0,2,4,6,11,15)$
      $D(A, B, C, D) = \Sigma(8,10,12,13,14)$
   ii. $F(A, B, C, D) = \Sigma(1.4.8.12,13,15)$
      $D(A, B, C, D) = \Sigma(3,7,11,14)$
   iii. $F(A, B, C, D) = \Sigma(0,1,2,3,7,8,10)$
      $D(A, B, C, D) = \Sigma(5,6,11,15)$
   iv. $F(A, B, C) = \Sigma(m_4, m_5, m_7)$
      $D(A, B, C) = \Sigma(m_0, m_2, m_3)$